INTERNATIONAL GCSE MATHEMATICS A · MODULAR · UNIT 2 HIGHER

Top 20 Questions for Unit 2

The questions that are coming

Twenty exam-style questions on the ideas in every Unit 2 paper — statistics, percentages, circle theorems, functions, vectors and calculus. One question per page, full solutions at the back.



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20	75	100%	1
QUESTIONS	MARKS	ORIGINAL QUESTIONS	QUESTION PER PAGE

THE TWENTY

1 Estimate of the mean
2 Cumulative frequency
3 Comparing box plots
4 Reverse percentage
5 Depreciation — find the rate
6 Standard form
7 Ratio that changes
8 Inverse proportion
9 Circle theorems
10 Similar triangles
11 Similar solids
12 Cone on a hemisphere
13 Simultaneous — line and curve
14 Functions
15 Maximum volume (calculus)
16 Vectors — straight line proof
17 Region on a graph
18 Arithmetic series
19 Change the subject
20 Trigonometric graph

BOTTOM LINE Do each question in full, on its page, without looking. Then mark it with the solutions at the back (page 22). Every question you needed the solution for: do it again tonight.

1 Estimate of the Mean

4 MARKS

Why it matters: a grouped table is usually Question 1 or 2.

The table gives information about the times, t minutes, that 40 students spent on homework.

Time (t minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	15
$20 < t \leq 30$	12
$30 < t \leq 40$	5

(a) Write down the modal class. **(1)**

(b) Work out an estimate for the mean time. **(3)**

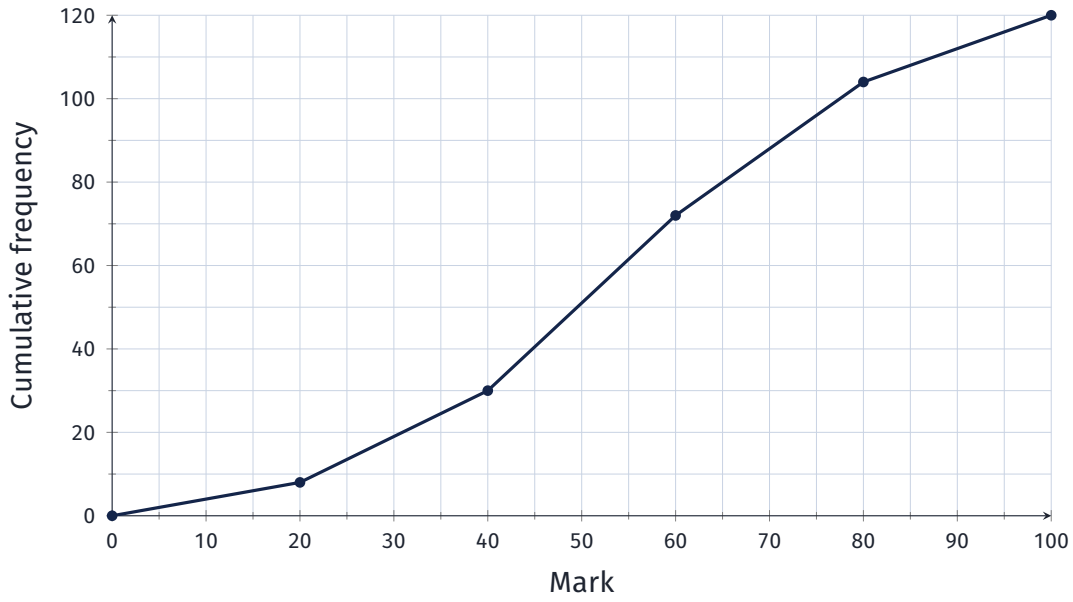
WORKING

2 Cumulative Frequency

5 MARKS

Why it matters: in every Unit 2 paper (Nov 2025 Q14, May 2025 R Q11).

The cumulative frequency graph shows the marks of 120 students in a test.



(a) Use the graph to find an estimate for the median mark. **(1)**

(b) Use the graph to find an estimate for the interquartile range. **(2)**

(c) The pass mark is 70. Work out an estimate for the number of students who passed. **(2)**

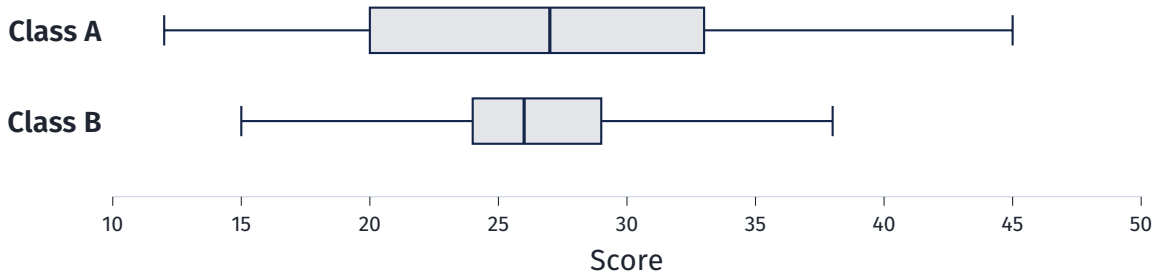
WORKING

3 Comparing Box Plots

3 MARKS

Why it matters: "compare" needs two comments, each with numbers.

The box plots show the scores of two classes, *A* and *B*, in a quiz.



(a) Work out the interquartile range for class *A*. **(1)**

(b) Compare the distributions of the scores of the two classes. **(2)**

WORKING

4 Reverse Percentage

3 MARKS

Why it matters: a “before the sale” question is in most papers.

In a sale, all prices are reduced by 15%.
The sale price of a coat is $91.80
Work out the price of the coat before the sale. **(3)**

WORKING

5 Depreciation — Find the Rate

3 MARKS

Why it matters: May 2025 R Q12 asked for the rate.

A car was bought for $24 000.
Its value decreased by the same percentage each year.
After 3 years its value was $14 739

Work out the percentage decrease each year. **(3)**

WORKING

6 Standard Form

3 MARKS

Why it matters: 2–3 quick marks every paper.

Work out each of these. Give your answers in standard form.

(a) $(4.2 \times 10^{5}) \times (3 \times 10^{-8})$ **(1)**

(b) $\dfrac{6 \times 10^{7}}{2.4 \times 10^{-3}}$ **(2)**

WORKING

7 Ratio That Changes

4 MARKS

Why it matters: a ratio that changes after something is added.

In a box, the ratio of red pens to blue pens is 3 : 5
12 red pens are added to the box.
The ratio of red pens to blue pens is now 9 : 10
Work out the number of blue pens in the box. **(4)**

WORKING

8 Inverse Proportion

3 MARKS

Why it matters: Nov 2025 Q17 — write the formula with k.

T is inversely proportional to the square root of m.
When $m = 16$, $T = 6$
Find the value of m when $T = 4$ **(3)**

WORKING

9 Circle Theorems

4 MARKS

Why it matters: every paper — and each step needs a reason.

A, *B* and *C* are points on a circle, centre *O*.
TA is the tangent to the circle at *A*.
Angle *TAB* = 58°

(a) Find the size of angle *ACB*. Give a reason. **(2)**

(b) Find the size of angle *OBA*. Give reasons. **(2)**

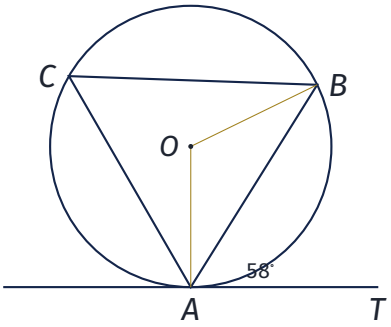


Diagram NOT accurately drawn

WORKING

10 Similar Triangles

3 MARKS

Why it matters: Nov 2025 Q5 — the scale factor from the WHOLE side.

ADB and *AEC* are straight lines. *DE* is parallel to *BC*.
AD = 6 cm, *DB* = 4 cm, *DE* = 9 cm, *AE* = 7.2 cm

(a) Work out the length of *BC*. **(2)**

(b) Work out the length of *EC*. **(1)**

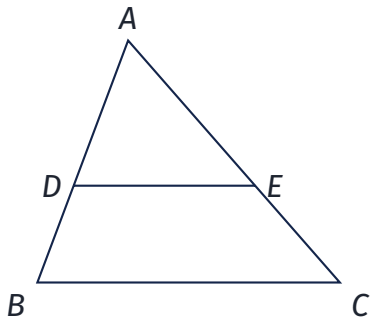


Diagram NOT accurately drawn

WORKING

11 Similar Solids

3 MARKS

Why it matters: Nov 2025 Q26 — length factor first.

Two bottles are mathematically similar. The smaller bottle has height 12 cm and the larger bottle has height 18 cm.

(a) The volume of the smaller bottle is 400 cm^3. Work out the volume of the larger bottle. **(2)**

(b) The surface area of the larger bottle is 540 cm^2. Work out the surface area of the smaller bottle. **(1)**

WORKING

12 Cone on a Hemisphere

4 MARKS

Why it matters: a compound solid is in every paper.

A solid is made from a cone on top of a hemisphere.
The cone and the hemisphere both have radius 5 cm.
The height of the cone is 12 cm.

(a) Work out the volume of the solid. Give your answer to 3 s.f. **(2)**

(b) Work out the total surface area of the solid. Give your answer to 3 s.f. **(2)**

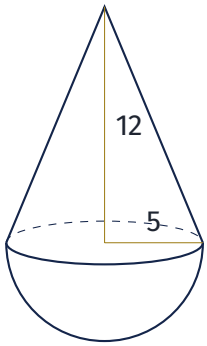


Diagram NOT accurately drawn

WORKING

13 Simultaneous — Line and Curve

5 MARKS

Why it matters: Nov 2025 Q20 — a 5-mark question every paper.

Solve the simultaneous equations $y = x + 1$ $x^2 + 2y^2 = 22$

Show clear algebraic working. **(5)**

WORKING

14 Functions

5 MARKS

Why it matters: Nov 2025 Q21 was 7 marks of functions.

$f(x) = 2x + 3$ $g(x) = x^2$ $h(x) = \dfrac{x + 1}{x - 2}$

(a) Find $fg(-2)$ **(1)**

(b) Find $h^{-1}(x)$ **(2)**

(c) Solve $gf(x) = 25$ **(2)**

WORKING

15 Maximum Volume

5 MARKS

Why it matters: May 2025 R Q21 — calculus on a real shape.

A square sheet of card has side 24 cm. A square of side x cm is cut from each corner and the sides are folded up to make an open box.

(a) Show that the volume of the box is $V = x(24 - 2x)^2$ **(1)**

(b) Find the value of x that gives the maximum volume, and the maximum volume. **(4)**

WORKING

16 Vectors — Straight-Line Proof

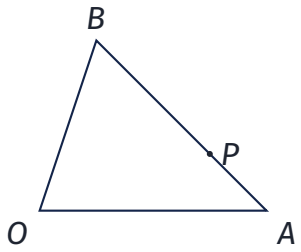
4 MARKS

Why it matters: Nov 2025 Q24 — the 6-mark vectors question.

$\vec{OA}$ = **a** and $\vec{OB}$ = **b**.
P is the point on AB such that $AP : PB = 1 : 2$
$\vec{OQ} = 2\mathbf{a} + \mathbf{b}$

(a) Find $\vec{OP}$ in terms of **a** and **b**. **(2)**

(b) Show that O, P and Q lie on a straight line. **(2)**



WORKING

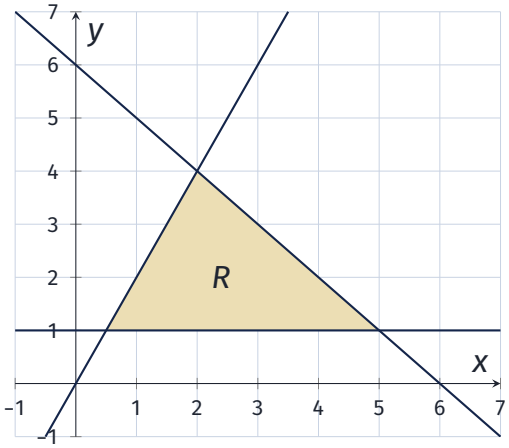
17 Region on a Graph

3 MARKS

Why it matters: "write down the three inequalities" — in both 2025 papers.

(3)

Write down the three inequalities that define the shaded region R.



WORKING

18 Arithmetic Series

4 MARKS

Why it matters: May 2025 R Q24 — 6 marks of series.

The 3rd term of an arithmetic series is 14. The 8th term is 34.
Work out the sum of the first 25 terms of the series. **(4)**

WORKING

19 Change the Subject

4 MARKS

Why it matters: May 2025 R Q14 — the letter appears twice.

Make k the subject of $p = \dfrac{3k + 2}{k - 5}$ **(4)**

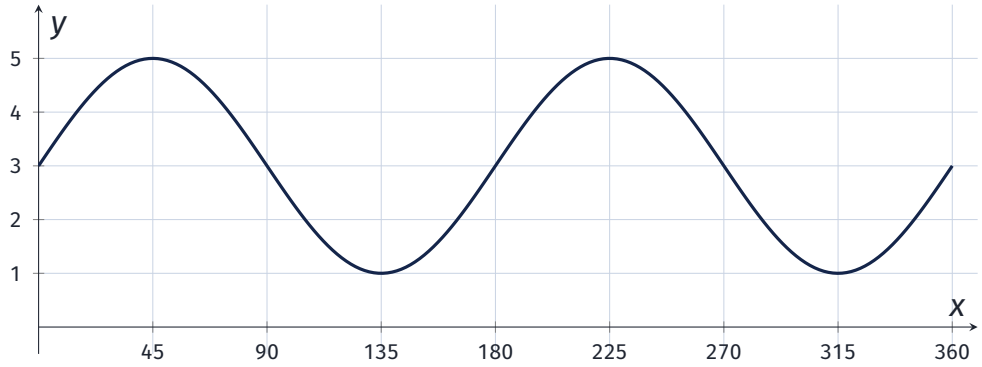
WORKING

20 Trigonometric Graph

3 MARKS

Why it matters: May 2025 R Q23 — read a, b, c from the graph.

The graph of $y = a\sin(bx^\circ) + c$ for $0 \le x \le 360$ is shown.



Find the value of a, of b and of c. **(3)**

WORKING

20 Full Solutions

Mark your work line by line. A method mark is for the right step even if the arithmetic slips — so check your steps, not only the answer.

1 Estimate of the mean

1 **MIDPOINTS × FREQUENCIES**
$5 \times 8 + 15 \times 15 + 25 \times 12 + 35 \times 5 = 40 + 225 + 300 + 175 = 740$

ANSWER (a) $10 < t \leq 20$ (b) $740 \div 40 = 18.5$ minutes

2 Cumulative frequency

2 **MEDIAN AT 60**
read across from 60: about 54

3 **QUARTILES AT 30 AND 90**
LQ = 40, UQ ≈ 71, IQR ≈ 31

4 **PASS MARK 70**
read up from 70: about 88 students scored 70 or less

ANSWER (a) 54 (b) 31 (c) 120 − 88 = 32

3 Comparing box plots

ANSWER (a) 33 − 20 = 13 (b) Class *A* has a higher median (27 > 26). Class *B* has a smaller IQR (5 < 13), so its scores are more consistent.

4 Reverse percentage

5 **SALE PRICE IS 85%**
$91.80 \div 0.85$

ANSWER $108

5 Depreciation — find the rate

6 **MULTIPLIER OVER 3 YEARS**
$14739 \div 24000 = 0.614125$

7 **ONE YEAR**
$\sqrt[3]{0.614125} = 0.85$

ANSWER 15%

6 Standard form

ANSWER (a) 1.26×10^{-2} (b) 2.5×10^{10}

7 Ratio that changes

8 **RED $3k$, BLUE $5k$**

$$\frac{3k + 12}{5k} = \frac{9}{10} \Rightarrow 30k + 120 = 45k \Rightarrow k = 8$$

ANSWER $5 \times 8 = 40$ blue pens

8 Inverse proportion

9 **FORMULA**

$$T = \frac{k}{\sqrt{m}} : 6 = \frac{k}{4} \Rightarrow k = 24$$

10 $T = 4$

$$\sqrt{m} = 6$$

ANSWER $m = 36$

9 Circle theorems

11 **(A)**

$ACB = 58^\circ$ — alternate segment theorem

12 **(B)**

$AOB = 2 \times 58 = 116^\circ$ (centre = 2× circumference); triangle OAB is isosceles ($OA = OB$, radii)

ANSWER (a) 58° (b) $(180 - 116) \div 2 = 32^\circ$

10 Similar triangles

13 **SCALE FACTOR FROM AD TO AB**

$10 \div 6 = \frac{5}{3}$

ANSWER (a) $BC = 9 \times \frac{5}{3} = 15$ cm (b) $AC = 12$, so $EC = 4.8$ cm

11 Similar solids

14 **LENGTH FACTOR**

$18 \div 12 = 1.5$

ANSWER (a) $400 \times 1.5^3 = 1350$ cm^3 (b) $540 \div 1.5^2 = 240$ cm^2

12 Cone on a hemisphere

15 **VOLUME**

$\frac{1}{3}\pi(25)(12) + \frac{2}{3}\pi(125) = 100\pi + \frac{250}{3}\pi$

16 **SURFACE**

slant $l = 13$: $\pi(5)(13) + 2\pi(25) = 65\pi + 50\pi = 115\pi$

ANSWER (a) 576 cm^3 (b) 361 cm^2

13 Simultaneous — line and curve

17 **SUBSTITUTE**

$x^2 + 2(x+1)^2 = 22 \Rightarrow 3x^2 + 4x - 20 = 0$

18 **FACTORISE**

$(3x + 10)(x - 2) = 0$

ANSWER $x = 2,\ y = 3$ or $x = -\frac{10}{3},\ y = -\frac{7}{3}$

14 Functions

19 **(A)**

$g(-2) = 4,\ f(4) = 11$

20 **(B)**

$y(x-2) = x + 1 \Rightarrow yx - x = 2y + 1 \Rightarrow x = \dfrac{2y+1}{y-1}$

21 **(C)**

$(2x+3)^2 = 25 \Rightarrow 2x + 3 = \pm 5$

ANSWER (a) 11 (b) $h^{-1}(x) = \dfrac{2x+1}{x-1}$ (c) $x = 1$ or $x = -4$

15 Maximum volume

22 **(A)**

base $(24 - 2x)$ by $(24 - 2x)$, height x

23 **DIFFERENTIATE**

$V = 576x - 96x^2 + 4x^3,\quad \dfrac{dV}{dx} = 576 - 192x + 12x^2$

24 **SET TO 0**

$12(x^2 - 16x + 48) = 0 \Rightarrow (x-4)(x-12) = 0$; $x = 12$ gives no box

ANSWER $x = 4,\quad V = 4 \times 16^2 = 1024\ \text{cm}^3$

16 Vectors — straight-line proof

25 **(A)**

$\vec{OP} = \mathbf{a} + \frac{1}{3}(\mathbf{b} - \mathbf{a}) = \frac{1}{3}(2\mathbf{a} + \mathbf{b})$

ANSWER (b) $\vec{OQ} = 2\mathbf{a} + \mathbf{b} = 3\vec{OP}$: parallel and both pass through O, so O, P, Q are on a straight line.

17 Region on a graph

ANSWER $y \geq 1$, $x + y \leq 6$, $y \leq 2x$

18 Arithmetic series

26 **TWO EQUATIONS**

$a + 2d = 14$, $a + 7d = 34 \Rightarrow d = 4$, $a = 6$

27 **SUM**

$S_{25} = \frac{25}{2}[12 + 24 \times 4]$

ANSWER 1350

19 Change the subject

28 **MULTIPLY OUT**

$pk - 5p = 3k + 2$

29 **k TERMS TOGETHER**

$pk - 3k = 5p + 2 \Rightarrow k(p - 3) = 5p + 2$

ANSWER $k = \dfrac{5p + 2}{p - 3}$

20 Trigonometric graph

30 **MIDDLE AND HEIGHT**

max 5, min 1: middle $c = 3$, height $a = 2$

31 **PERIOD**

one full wave in 180°, so $b = 360 \div 180 = 2$

ANSWER $a = 2$, $b = 2$, $c = 3$